



VELALAR COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)

Approved by AICTE, affiliated to Anna University, and Accredited by NAAC with 'A+' Grade



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Accredited by NBA)

AN INTERNSHIP REPORT

Submitted by

ELANNGO M
(22CSR031)

in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING IN
COMPUTER SCIENCE AND ENGINEERING

MAY 2026



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BONAFIDE CERTIFICATE

This is to certify that the Internship Report entitled “**REACT NATIVE APP**”, in partial fulfillment of the requirements for the award of the Degree of BACHELOR OF ENGINEERING is a record of original training undergone by **ELANNGO M (22CSR031)** during the period from (23.06.2025) to (01.05.2026) of her study in the department of COMPUTER SCIENCE AND ENGINEERING, VELALAR COLLEGE OF ENGINEERING AND TECHNOLOGY and the report has not formed the basis for the award of any Degree/Fellowship or other similar title to any candidate of any University.

FACULTY IN-CHARGE

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Assistant Professor,

Department of Computer Science and Engineering,

Velalar College of Engineering and Technology,

Thindal, Erode-638012.

HEAD OF THE DEPARTMENT

Dr. S. JABEEN BEGUM, M.E., Ph.D.,

Professor and Head,

Department of Computer Science and Engineering,

Velalar College of Engineering and Technology,

Thindal, Erode-638012.

CERTIFICATE FROM THE INDUSTRY



Personal and Confidential

23rd June 2025

To
Elanngo M,
S/O Madheswaran,
104/3 Ponlakmi Illam,
Udhayam Garden,
Erode, TamilNadu – 638107.

Sub: Offer Letter for Training Program

Dear Elanngo,

We are pleased to offer you an opportunity to work with our organization and your Training starts from **23rd June 2025** with a stipend of **Rs.11,100 Per Month**.

This period may be extended or reduced depending upon your performance, application, attendance and bearing. The management will assign your work timings. Our normal working hours are Monday to Saturday from **9AM to 6 PM**.

If your performance is found to be satisfactory during such period, you may confirm full time employment during or after the completion of training program.

You shall adhere to all company policies/procedures and shall not reveal/disclose/publish in any manner any of the information, details, processes, and policies of the company to any third party in any form (oral, written, social networking sites etc) without prior approval of the management. You should be aware that the training provided to you with the Company is liable to be terminated at any time with or without notice, with or without good cause or for any or no cause, at the company's option.

You will be required to sign an NDA on the day of joining.

We wish all the best for your internship program with us.

Sincerely,

For BI3 Technologies India Pvt Ltd,

Accepted by

Sincerely,

Anitha M

HR Lead



BI3 Technologies India Pvt Ltd

CIN: U72200TN2021FTC148693

www.bi3technologies.com | 044 45508333

Regd. Office: No. 75, 1st floor, 3rd Avenue, Anna Nagar East, Chennai, Tamil Nadu 600102

DECLARATION

I, **ELANNGO M**, hereby declare that the Internship Report, entitled “**REACT NATIVE APP**”, submitted to the **Velalar College of Engineering and Technology** in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Engineering** is a record of original training undergone by me during the period (**Jun-May 2026**) and it has not formed the basis for the award of any Degree/Fellowship or other similar title to any candidate of any University.

Place:

Signature of the Student

Date:

ACKNOWLEDGEMENT

On the occasion of successfully completing my internship project, I wish to express my heartfelt gratitude to the honorable Secretary and Correspondent of the Vellalar Educational Trust, **Thiru. S. D. CHANDRASEKAR B.A.**, for providing the necessary infrastructure and support to accomplish for my internship.

I am deeply thankful to my Principal, **Dr. M. JAYARAMAN Ph.D.**, of Velalar College of Engineering and Technology, for his continuous support and encouragement throughout my academic journey.

I would also like to express my sincere appreciation to my Dean, **Prof. P. JAYACHANDAR M.E.**, for granting me the opportunity to undertake this internship and for his constant motivation.

My deepest gratitude goes to my Head of the Department, **Dr. S. JABEEN BEGUM M.E., Ph.D.**, whose insightful guidance and encouragement helped me navigate through the various stages of this project.

A special thanks to my Internship Coordinator, **Ms. R. DHARANI B.E., M.E.**, Assistant Professor, and my Internship Supervisor, **Mrs. B. MOHANAPRIYA B.E., M.E.**, Assistant Professor, for their technical guidance, timely suggestions, and unwavering support, which were instrumental in the successful completion of my internship project.

I also wish to extend my thanks to all the teaching and non-teaching staff members of the Computer Science and Engineering Department for their assistance.

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CHAPTER 1

ABSTRACT

DEVELOPMENT OF A REACT NATIVE APP

This report presents the development of a cross-platform mobile application built using React Native, designed to provide real-time factory analytics through a unified and user-friendly interface. The application focuses on delivering actionable insights such as employee attendance, customer congestion levels, and operational trends to support efficient monitoring and decision-making.

The core functionality is powered by a visual analytics pipeline that processes incoming data from multiple sources and transforms it into meaningful insights. These insights are presented through intuitive dashboards within the app, enabling users to easily interpret complex data through clear visualizations. The system emphasizes a clean and minimal UI design, ensuring a seamless and engaging user experience.

The application architecture follows a modular approach, separating data processing, analytics generation, and UI rendering to enhance scalability, maintainability, and performance. Real-time updates ensure that users always have access to the latest operational data, making the application suitable for dynamic industrial environments.

By leveraging React Native, the application supports seamless deployment across both Android and iOS platforms using native components, reducing development overhead while maintaining near-native performance. This project demonstrates effective implementation of cross-platform development, real-time data visualization, and intuitive UI/UX design for industrial analytics applications.

Keywords:

React Native, Cross-Platform Development, Factory Analytics, Attendance Tracking, Customer Congestion Analysis, Data Visualization, Mobile Dashboard, Real-Time Data, UI/UX Design, Embedded Analytics

CHAPTER 2

INTRODUCTION ABOUT THE COMPANY

BI3 TECHNOLOGIES is a forward-thinking data and analytics company dedicated to transforming raw data into meaningful insights and actionable business value. With a strong focus on innovation, scalability, and security, the company empowers organizations to harness the full potential of their data through modern cloud-based platforms and advanced analytics solutions.

At the core of BI3 TECHNOLOGIES's operations is its comprehensive service portfolio spanning across data integration, platform engineering, and intelligent analytics. The company specializes in building real-time and IoT-driven data ecosystems, enabling businesses to gain instant visibility into operations, optimize processes, and unlock new growth opportunities. Their expertise in delivering scalable and cost-effective data platforms ensures seamless handling of large-scale data across diverse environments. BI3 TECHNOLOGIES also excels in data migration and modernization, helping organizations transition from legacy systems to modern data architectures with minimal disruption. Through managed services and continuous support, the company ensures stability, reliability, and long-term performance of data platforms.

Beyond infrastructure, BI3 TECHNOLOGIES places a strong emphasis on data intelligence. Their capabilities include defining robust data strategies, implementing governance frameworks, and ensuring high data quality and consistency through master data management practices. This enables organizations to build a trusted and unified view of their business data. On the innovation front, BI3 TECHNOLOGIES leverages cutting-edge technologies such as machine learning, artificial intelligence, and computer vision to deliver advanced analytics solutions. From predictive insights and process automation to interactive data visualizations and video analytics, the company enables data-driven decision-making across industries.

BI3 TECHNOLOGIES continues to build a reputation for delivery excellence, operational efficiency, and customer success, helping organizations transform their data into a strategic asset and maintain a competitive edge in an increasingly data-driven world.

CHAPTER 3

COMPANY PROFILE

SURESOFT SYSTEM'S PVT LTD

BI3 TECHNOLOGIES is a leading data and analytics solutions provider that delivers modern, scalable, and intelligent data-driven systems for enterprises across industries. The company specializes in transforming complex data into actionable insights through advanced analytics, cloud-based platforms, and intuitive visualization tools. BI3 focuses on enabling organizations to make informed decisions by integrating cutting-edge technologies with user-centric solutions.

Establishment:

BI3 TECHNOLOGIES has established itself as a trusted partner in the data and analytics domain, with a strong presence in delivering enterprise-grade solutions globally. Founded on the principles of unity, diversity, and collaboration, the company brings together expertise from multiple cultural and technical backgrounds to create impactful and innovative data solutions.

Key Offerings:

- **Data Platform Delivery:** Design and implementation of scalable, secure, and cost-effective data platforms using modern cloud technologies.
- **Real-Time & IoT Integration:** Enabling organizations to leverage real-time data streams and IoT systems for enhanced operational visibility and optimization.
- **Data Migration & Modernization:** Seamless migration of legacy data systems and reporting tools to modern, efficient architectures.
- **Managed Services & Operations:** Continuous monitoring, maintenance, and support for data platforms ensuring reliability and performance.
- **Data Strategy & Architecture:** Aligning data initiatives with business goals through structured strategies and architectural frameworks.
- **Data Governance & Quality Management:** Ensuring data integrity, security, compliance, and consistency across enterprise systems.
- **Machine Learning & AI:** Leveraging advanced analytics, predictive modeling, and AI techniques to derive insights and automate decision-making.
- **Data Visualization & Reporting:** Development of interactive dashboards and reports for

effective communication of insights.

- **Process Automation:** Implementation of workflow and robotic process automation to improve efficiency and reduce manual effort.
- **Video Analytics:** Application of computer vision and AI techniques to extract insights from video data.

Company Values:

- Data-Driven Innovation and continuous improvement
- Excellence in delivery, scalability, and performance
- Customer-Centric approach with tailored analytics solutions
- Strong focus on data security, governance, and compliance
- Collaboration through diversity and global expertise

Company Information:

- **Location:** No 75, 1st Floor, M Block 3rd Avenue, Anna Nagar East, Chennai – 60010, Tamil Nadu, India.
- **Contact:** Ph: 044 45508333, contacts@bi3technologies.com

Specializations:

Data Engineering, Cloud Data Platforms, Data Governance, Machine Learning, AI Solutions, Data Visualization, Process Automation, IoT Analytics.

Work Culture:

BI3 TECHNOLOGIES promotes a collaborative and innovation-driven work environment where diverse teams come together to solve complex data challenges. The organization emphasizes continuous learning, knowledge sharing, and hands-on project involvement. Employees and interns work on real-world data problems, gaining exposure to modern tools, cloud platforms, and advanced analytics practices while contributing to impactful business solutions.

Technology Stack:

BI3 TECHNOLOGIES leverages a modern and flexible technology ecosystem that includes:

- Data Platforms: Snowflake, Azure Synapse, Amazon Redshift
- Visualization Tools: Power BI, Tableau, Qlik
- Programming & Analytics: Python, SQL, Machine Learning frameworks
- Automation Tools: Power Automate, RPA solutions
- AI & Video Analytics: Computer Vision, ML-based analytics

CHAPTER 4

INTERNSHIP OVERVIEW- MONTHLY REPORT

4.1 DOMAIN OF THE TRAINING

Data Analytics Application Development and Real-Time Visualization Systems

Data Analytics Application Development is a specialized domain within software engineering that focuses on transforming raw data into meaningful insights through processing, analysis, and visualization. Unlike traditional application development, this domain emphasizes handling large-scale, real-time, and heterogeneous data while ensuring scalability, performance, and usability. It is widely applicable across industries such as manufacturing, retail, healthcare, and smart infrastructure, where data-driven decision-making plays a crucial role.

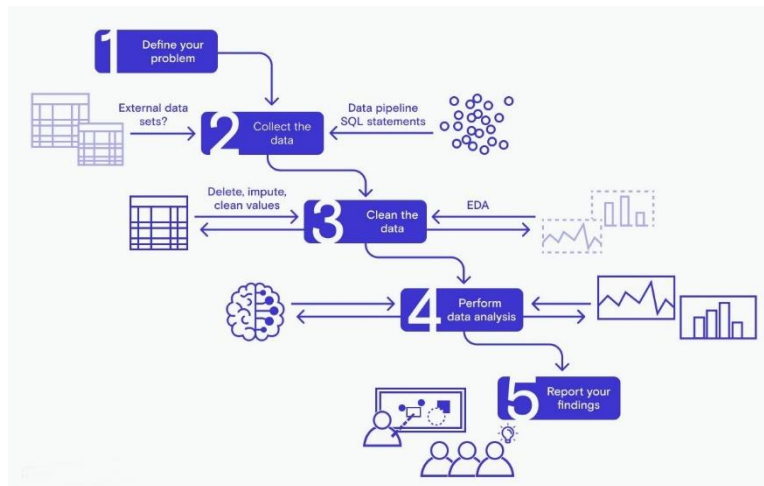


FIG NO: 4.1 KEY COMPONENTS OF DATA ANALYTICS

The training domain encompasses the following core components:

Cross-Platform Mobile Application Development (React Native):

A key aspect of the training involved developing mobile applications using React Native. This enabled the creation of a single codebase that can be deployed across both Android and iOS platforms while utilizing native components for performance and user experience. I gained hands-on experience in building responsive user interfaces, managing component lifecycles, and integrating APIs to display real-time analytics in a clean and intuitive format.

Real-Time Data Integration and Processing:

The application focused on consuming and processing real time data streams related to factory operations such as employee attendance and customer congestion. I learned how to handle dynamic data inputs, ensure efficient data flow between backend systems and the mobile application, and maintain synchronization to reflect up to date information for end users.

Data Visualization and Dashboard Design:

A significant part of the project involved transforming complex datasets into meaningful visual representations. I worked on designing interactive dashboards that present analytics through charts, graphs, and summarized metrics. This helped in understanding how effective visualization improves decision-making by making data more accessible and interpretable.

Analytics Pipeline Understanding:

The system is supported by a visual analytics pipeline that processes raw data into structured insights. I gained exposure to how data flows from collection sources through processing layers and is ultimately delivered to the application. This provided a foundational understanding of data pipelines, including aggregation, transformation, and delivery of insights.

User Interface and User Experience (UI/UX) Design:

The application emphasizes a minimal and user-friendly design to ensure ease of use. I contributed to designing clean layouts, intuitive navigation, and responsive elements that enhance usability. This experience reinforced the importance of balancing functionality with simplicity in analytics-driven applications.

Attendance and Congestion Analytics Systems:

The project included modules for monitoring employee attendance and analyzing customer congestion patterns. I worked on integrating these analytics into the application, enabling real-time tracking and visualization. This helped in understanding practical use cases of data analytics in operational environments.

API Integration and Data Handling:

To connect the mobile application with backend services, I implemented API integrations for fetching and updating data. This involved handling asynchronous requests, managing state efficiently, and ensuring reliable data rendering within the application.

Modular Application Architecture:

Throughout the development process, I followed a modular approach by separating the application into distinct layers such as data handling, business logic, and UI components. This improved code maintainability, scalability, and reusability—essential for modern application development.

Real-Time Analytics Application Design:

The overall training provided a strong foundation in designing applications that handle real-time data and deliver actionable insights. I developed an understanding of how to build responsive systems that adapt to continuous data updates while maintaining performance and usability.

4.2 AIM & OBJECTIVES OF TRAINING

To develop a scalable, cross-platform mobile application for real-time factory analytics, enabling efficient visualization of operational data such as attendance and customer congestion through a clean and intuitive user interface using React Native.

This overarching goal was supported by several specific objectives:

1. To design and develop a cross-platform mobile application using React Native that works seamlessly on both Android and iOS devices.
2. To integrate real-time data sources and APIs for fetching and displaying factory analytics such as employee attendance and customer congestion.
3. To understand and implement data flow from analytics pipelines to the mobile application for accurate and timely data representation.
4. To design interactive dashboards and visual components that effectively communicate complex data insights through charts and summaries.
5. To ensure a clean, minimal, and user-friendly UI/UX that enhances usability and improves user interaction with analytics data.
6. To implement efficient state management and asynchronous data handling
7. To structure the application using a modular architecture, separating data handling, business logic, and UI components for better maintainability and scalability.
8. To test and validate the application using real-time data scenarios, ensuring reliability, responsiveness, and performance.

4.3 SCOPE OF TRAINING

The scope of the data analytics and mobile application development project encompasses:

1. Technical Foundations:

- o Development of cross-platform mobile applications using React Native.
- o Integration of frontend applications with backend data services through APIs.
- o Handling asynchronous operations and real-time data updates within the application.
- o Implementation of modular programming practices and component-based architecture.
- o Understanding of cloud-based data platforms and analytics ecosystems.

2. Real-Time Data Processing:

- o Integration of real-time data streams related to factory operations such as attendance and customer congestion.
- o Ensuring efficient data synchronization between backend systems and the mobile application.
- o Handling dynamic data updates and maintaining application responsiveness.

3. Data Visualization and Dashboard Development:

- o Designing interactive dashboards using charts, graphs, and summary metrics.
- o Transforming raw data into meaningful visual insights for better decision-making.
- o Ensuring clarity, readability, and usability in presenting analytical data.

4. Analytics Integration:

- o Understanding the flow of data through analytics pipelines from data sources to visualization layers.
- o Implementing logic to display processed insights within the application.
- o Supporting data-driven features such as trend analysis and operational monitoring.

5. Attendance and Congestion Monitoring Systems:

- o Developing modules to track employee attendance data in real time.

- o Implementing customer congestion analysis for operational insights.
- o Visual representation of these metrics for quick interpretation and decision support.

6.API Integration & Data Handling:

- o Connecting the mobile application with backend services using RESTful APIs.
- o Managing data fetching, error handling, and state updates efficiently.
- o Ensuring reliable and consistent data rendering within the UI.

7.UI/UX Design and Optimization:

- o Designing a clean, minimal, and intuitive user interface.
- o Enhancing user experience through responsive layouts and smooth navigation.
- o Optimizing performance for seamless interaction with real-time data.

8.Software Design & Architecture:

- o Adopting a modular and layered architecture separating UI, logic, and data layers.
- o Ensuring scalability, maintainability, and reusability of code components.
- o Following best practices in application structuring and version control.

4.4 TRAINING DETAILS

MONTH 1: SYSTEM UNDERSTANDING AND DEVELOPMENT SETUP

Overview:

The first month focused on understanding the project requirements, data analytics concepts, and setting up the development environment. I was introduced to the fundamentals of data-driven application development and cross-platform mobile technologies.

Key Learning Areas:

- 1 Functional analysis of factory analytics (attendance, congestion monitoring)
- 2 Introduction to data analytics pipelines and real-time data flow
- 3 Familiarization with React Native and mobile development concepts
- 4 Environment setup: IDEs, dependencies, and project structure

Practical Tasks:

- 1 Prepared application flow diagrams and UI wireframes
- 2 Set up development environment and configured project dependencies
- 3 Created basic React Native screens and navigation structure
- 4 Tested simple API integrations

Skills Developed:

- 1 Fundamentals of React Native development
- 2 Understanding of data flow in analytics systems
- 3 Basic UI design and component structuring

MONTH 2: CORE APPLICATION DEVELOPMENT AND API INTEGRATION**Overview:**

The second month focused on building the core structure of the mobile application and integrating backend APIs for data retrieval.

Key Learning Areas:

- 1 Decoding NEC IR protocol
- 2 Interrupt-driven programming in embedded systems
- 3 Mapping IR codes to channel numbers
- 4 State-machine-based design for remote control inputs

Practical Tasks:

- o Integrated APIs for fetching attendance and congestion data
- o Implemented state management for dynamic data updates
- o Developed reusable UI components
- o Handled loading states and error responses

Skills Developed:

- o API integration and data handling
- o Asynchronous programming concepts
- o Component-based architecture design

MONTH 3: DATA VISUALIZATION AND DASHBOARD DEVELOPMENT

Overview:

This month focused on transforming raw data into meaningful insights through interactive dashboards and visual components.

Key Learning Areas:

- 1 Data visualization principles and best practices
- 2 Designing dashboards for analytics applications
- 3 Representation of real-time data using charts and graphs

Practical Tasks:

- o Developed dashboards for attendance and congestion analytics
- o Integrated chart libraries for data visualization
- o Designed summary views and key performance indicators (KPIs)
- o Optimized layout for readability and clarity

Skills Developed:

- o Data visualization techniques
- o Dashboard design and analytics presentation
- o UI optimization for complex data

MONTH 4: REAL-TIME DATA HANDLING AND FEATURE ENHANCEMENT

Overview:

The fourth month focused on handling real-time updates and enhancing application features for better performance and usability.

Key Learning Areas:

1. Real-time data synchronization techniques
2. Performance optimization in mobile applications
3. Enhancing analytics features for better insights

Practical Tasks:

- o Implemented real-time data updates in dashboards
- o Optimized API calls and rendering performance
- o Enhanced analytics modules with additional insights
- o Improved responsiveness across devices

Skills Developed:

- o Real-time data handling
- o Application performance optimization
- o Advanced analytics integration

MONTH 5: TESTING, DEBUGGING, AND OPTIMIZATION**Overview:**

This phase focused on ensuring application stability, accuracy, and performance through systematic testing and debugging.

Key Learning Areas:

1. Testing strategies for mobile applications
2. Debugging techniques and performance analysis
3. Handling edge cases and error scenarios

Practical Tasks:

- o Tested application with real-time and simulated data
- o Debugged UI issues and API inconsistencies
- o Implemented error handling and fallback mechanisms
- o Optimized application performance and responsiveness

Skills Developed:

- o Application debugging and troubleshooting
- o Error handling and system stability improvement
- o Performance tuning techniques



FIG NO: 4.4 REACT NATIVE APP BUILD CONSOLE OUTPUT

MONTH 6: FINALIZATION, DOCUMENTATION, AND DEPLOYMENT

Overview:

The final month focused on polishing the application, preparing documentation, and demonstrating the completed system.

Key Learning Areas:

1. Documentation standards for software projects
2. Application deployment processes
3. Demonstration and presentation techniques

Practical Tasks:

- o Prepared technical and user documentation
- o Finalized UI/UX improvements

- o Demonstrated application features and workflow
- o Proposed future enhancements such as advanced analytics and automation features

Skills Developed:

- o Technical documentation and reporting
- o Project presentation skills
- o End-to-end application development lifecycle understanding

CHAPTER 5

CONCLUSION

My internship at BI3 TECHNOLOGIES has been a highly enriching and transformative experience, significantly contributing to both my technical growth and professional development. Specializing in data analytics and mobile application development, this internship provided me with a strong understanding of how modern data-driven solutions are designed, developed, and deployed in real-world business environments. Working with cross-platform technologies, I gained hands-on experience in building applications that transform complex data into meaningful insights.

One of the key highlights of this internship was the development of a real-time analytics mobile application using React Native. Through this project, I learned how to integrate backend data sources with a mobile frontend, handle real-time data updates, and present insights through intuitive dashboards. This experience enhanced my understanding of data visualization, user interface design, and efficient data handling techniques, which are critical in analytics driven applications.

Additionally, I gained exposure to the complete lifecycle of a data analytics solution from understanding business requirements and data flow to implementing features and optimizing performance. I worked with APIs, managed asynchronous data operations, and structured applications using modular design principles. This helped me appreciate the importance of scalability, maintainability, and performance in modern software systems.

The internship also provided valuable insights into working within a professional and collaborative environment. Participating in team discussions, iterative development processes, and feedback cycles helped me improve my communication skills and ability to approach problems from a practical perspective. I developed a deeper understanding of coding standards, documentation practices, and structured development workflows followed in industry settings.

Another important aspect of this experience was learning the significance of data accuracy,

reliability, and real-time responsiveness. I understood how critical it is to design systems that not only present data effectively but also ensure consistency and correctness, especially in operational and decision-making contexts.

Overall, my internship at BI3 TECHNOLOGIES has strengthened my interest in data analytics and application development. It enabled me to bridge the gap between theoretical knowledge and practical implementation while equipping me with essential skills in cross-platform development, real-time data processing, and analytics visualization. This experience has built a solid foundation for my future career and has given me the confidence to work on complex, data driven software solutions in a professional environment.